

rest of the market will usually learn of this mispricing and adjust their market quotes to reflect the proper pricing at the open the next day.

Future Price Benchmark

Index managers often select the closing price to be their benchmark because this is the price that will be used to value the fund. Any transaction that is different than the closing price on the day of the trade will cause the actual value of the fund to differ from the index benchmark price thus causing tracking error. In order to avoid incremental tracking error and potential subpar performance, index managers often seek to achieve the closing price on the day.

An interesting aspect of using a future price as the benchmark is that your performance will look better than it actually is because the future price will also include the permanent impact of the order. So while permanent impact will have an adverse effect on the arrival price or historical price benchmark, it will not have any effect on a future price benchmark (from a cost perspective). Investors are expected to perform better against a future price benchmark by the amount of the permanent impact than they will against the arrival price.

The future cost derivation is explained as follows:

$$E_0[\text{Future Cost}] = (\bar{P} - E(P_n))$$

The E_0 notation is used to denote that this is the expected cost at the beginning of trading. Thus,

$$\begin{aligned} E(P_n) &= P_0 + (1 - b_1) \cdot I' \\ \bar{P} &= P_0 + (b_1 \cdot I' \cdot \alpha) + (1 - b_1) \cdot I' \end{aligned}$$

The expected future cost is:

$$E_0[\bar{P} - E(P_n)] = (b_1 \cdot I' \cdot \alpha) = \text{Temporary Market Impact}$$

And we have,

$$E_0[\text{Future Cost}] = (b_1 \cdot I' \cdot \alpha) \quad (8.11)$$

Notice that the future cost function only consists of temporary market impact. As stated, this is because permanent impact is reflected in the future price. So the future cost function that only includes temporary impact will be lower than the arrival cost function that includes both temporary and permanent impact.